Fiscal Year:	FY 2007	**Task Last Updated:**	FY 02/09/2008
PI Name:	Kuzmich, Alex Ph.D.		
Project Title:	Entangled atomic clock		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Atlanta	**State:**	GA
Zip Code:	30332-0100	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	04/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	7	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 09/30/2007 per PI (7/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kennedy, T.A.Brian (School of Physics)		
Grant/Contract No.:	GSN 4516		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A four-year program of experimental and theoretical research on development of entanglement-enhanced atomic clock will be undertaken. Atomic entanglement will be generated via collective QND measurements of populations of the two levels involved in the atomic clock transition. In the first phase of the research, experimental and theoretical investigations will be done with a cold sample of atoms typical of present-day atomic clocks. In the second stage, we will investigate BEC-based entangled atomic clock. For the latter case we will pursue theoretical and experimental investigations of the issues of stability of spatial modes and the impact of the collisional frequency shift on the entanglement-enhanced performance of such a clock. We intend to experimentally implement a quantum non-demolition interaction between our atomic ensemble and a light field. Measurement of the light field amounts to a quantum non-demolition measurement of the atoms, creating an entangled state. The essential new idea we propose is the use of a single off-resonant light field which when correctly tuned with respect to the hyperfine levels measures the population difference of the levels levels $|1\rangle $ and $|2\rangle $ associated with the clock transition. Moreover, under these conditions the interaction conserves the population difference of the atomic sample, a vital element in our experimental measurement strategy. Another objective of the research program is to investigate methods which allow cancellation of classical noise sources, such as those associated with distribution of coupling strengths seen by an atomic ensemble in a spatially varying field mode. If uncorrected, such noise could entirely mask entanglement, precluding the possible enhancement in accuracy of an atomic clock. We propose the use of ``matched" observables, which enable large amounts of entanglement to be extracted under these circumstances of apparently weak entanglement. We will investigate the concept of matched observables in the contexts of both cold non-degenerate and BEC-based atomic clocks. The proposed work will significantly impact possible future microgravity atomic clock experiments. Those experiments will be attempting to reach higher accuracy than ground-based atomic clocks. One avenue to even higher accuracy is to increase the number of atoms and/or interaction time. However, this avenue may have already been pushed close to its limits. The development of an entanglement-enhanced atomic clock would represent a major breakthrough and pave the way for further improvements. Measurement-induced entanglement on the atomic clock transition may very well become an integral part of the space-based atomic clock of the future.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research on quantum repeaters relates to a potentially secure information transfer system using quantum mechanics.
Task Progress:	Our major research effort has been building a long-distance quantum communication system. We have laid the groundwork for implementation of long-distance quantum communication. Particularly, we proposed a practical quantum telecommunication scheme and realized its critical elements in our laboratory. We have implemented critical elements of quantum networks - collective atomic quantum bits, quantum state transfer between matter and light, entanglement of atomic and photonic qubits, long lived quantum mmemories, entangled remote atomic qubits, entanglement of a telecom wavelength photon and of a storable photon. Our work should impact the fields of communication and computation, as it provides technical capabilities that have not been previously available. Graduate and undergraduate students have been trained in the technologically important area of optical science. Students and postdocs have acquired valuable laboratory skills, and presented their findings to wide audiencies at major national and international conferences.
Bibliography Type:	Description: (Last Updated: 02/13/2008)
Articles in Peer-reviewed Journals	Lan SY, Jenkins SD, Chaneliere T, Matsukevich DN, Campbell CJ, Zhao R, Kennedy TA, Kuzmich A. "Dual species matter qubit entangled with light." Phys Rev Lett. 2007 Mar 23;98(12):123602. PMID: 17501122 , Mar-2007
Articles in Peer-reviewed Journals	Collins OA, Jenkins SD, Kuzmich A, Kennedy TA. "Multiplexed memory-insensitive quantum repeaters." Phys Rev Lett. 2007 Feb 9;98(6):060502. PMID: 17358924 , Feb-2007
Articles in Peer-reviewed Journals	Chaneliere T, Matsukevich DN, Jenkins SD, Lan SY, Zhao R, Kennedy TA, Kuzmich A. "Quantum interference of electromagnetic fields from remote quantum memories" Phys Rev Lett. 2007 Mar 16;98(11):113602. PMID: 17501054 , Mar-2007
Articles in Peer-reviewed Journals	Jenkins SD, Matsukevich DN, Chaneliere T, Lan S-Y, Kennedy TAB, Kuzmich A. "Quantum telecommunication with atomic ensembles." Journal of the Optical Society of America. B, Optical physics. 2007 Feb;24(2):316-23. http://www.opticsinfobase.org/abstract.cfm?URI=josab-24-2-316 , Feb-2007